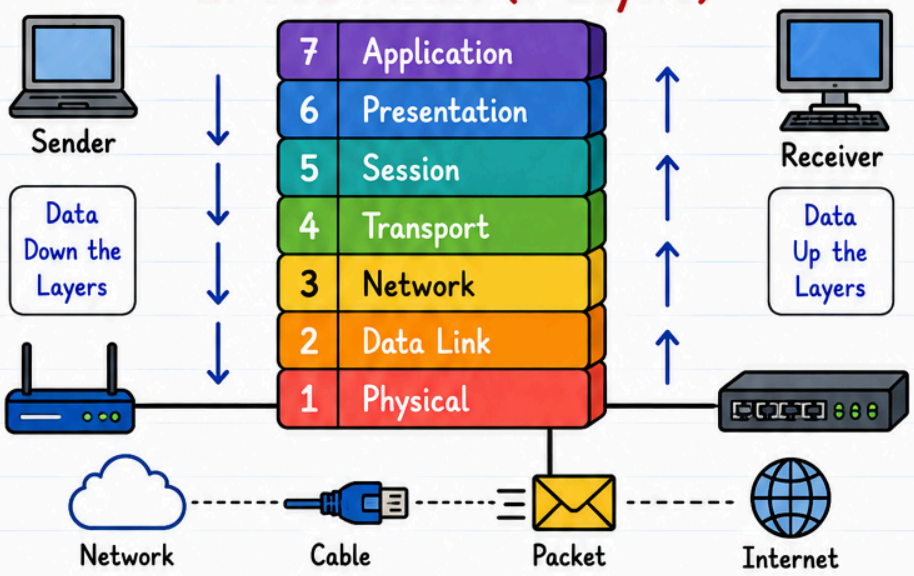
Networking for Cloud & DevOps

1. OSI Model (7 Layers)



OSI Model is a standard 7-layer reference model used to understand how network communication works. It helps explain where protocols and devices operate in a network.

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1. Purpose

- Standard model for understanding network communication.
- Breaks communication into 7 logical layers.
- Helps in troubleshooting and protocol mapping.

2. 7 Layers

7	Application	HTTP, HTTPS, DNS
6	Presentation	SSL/TLS, Encryption
5	Session	Session Management
4	Transport	TCP, UDP
3	Network	IP, Routing
2	Data Link	MAC Address, Switch
1	Physical	Cable, Fiber, Hub

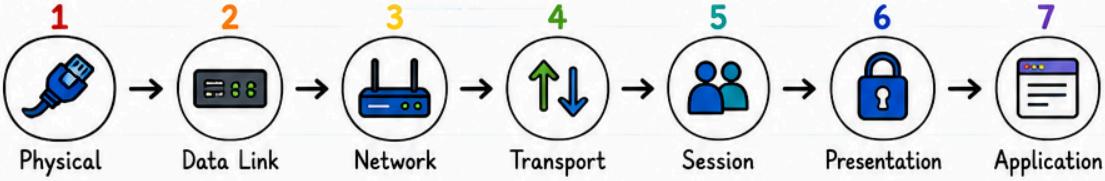
3. Remember

All People Seem To Need Data Processing

A	P	S	T	N	D	P
Application	Presentation	Session	Transport	Network	Data Link	Physical

4. Why Important in DevOps?

- Helps understand traffic flow and failures.
- Useful in cloud, Kubernetes, and security troubleshooting.
- Layer 3 = IP/Routing, Layer 4 = TCP/UDP, Layer 7 = HTTP/HTTPS.



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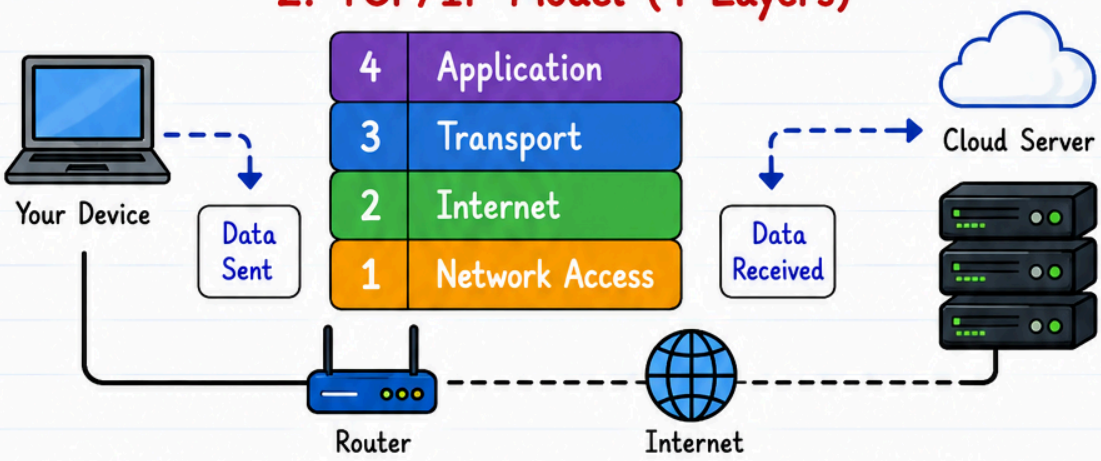
2. TCP/IP Model (4 Layers)



The TCP/IP model is the practical networking model used on the Internet. It groups communication into 4 layers and defines how real devices exchange data online.

1. Purpose

- Practical model used for Internet communication.
- Groups protocols into 4 layers.
- Explains how data moves across networks.

2. 4 Layers

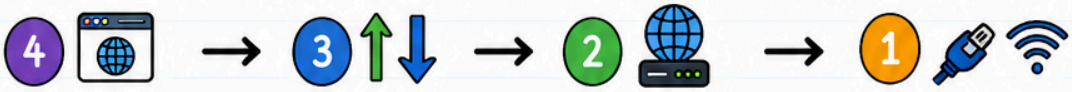
4		Application	HTTP, DNS, FTP, SMTP
3		Transport	TCP, UDP
2		Internet	IP, ICMP
1		Network Access	Ethernet, Wi-Fi

3. OSI vs TCP/IP

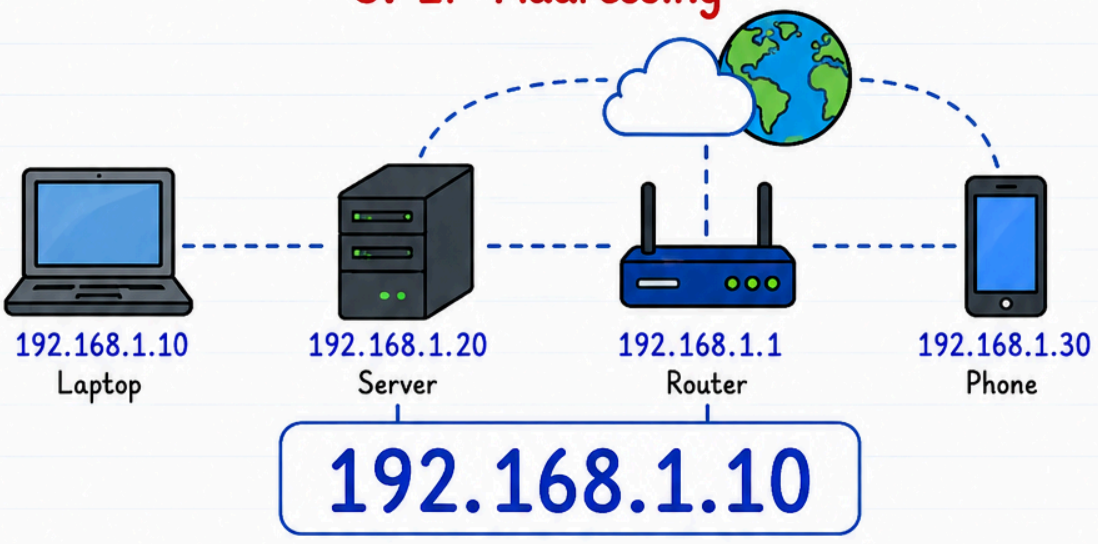
TCP/IP Model	OSI Model
Application	OSI Layers 5, 6, 7
Transport	OSI Layer 4
Internet	OSI Layer 3
Network Access	OSI Layers 1, 2

4. Key Takeaway

- TCP/IP is the foundation of the modern Internet.
- TCP provides reliable delivery.
- UDP is lightweight and fast.
- IP handles addressing and routing.



3. IP Addressing



An IP address is a unique address assigned to a device on a network. It allows devices to identify each other and communicate across local networks and the Internet.

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1. What is IP Addressing?

- Unique address assigned to every device.
- Helps identify source and destination.
- Essential for communication.

2. Example

192.168.1.10

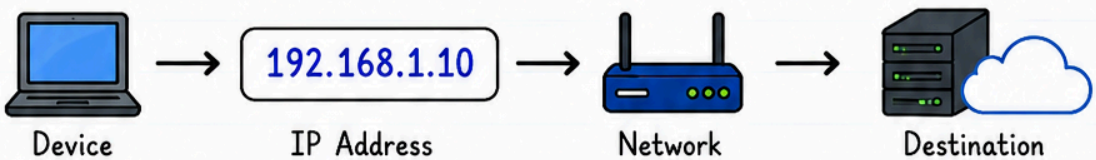
Private IPv4 example.

3. Types

IPv4	IPv6
32-bit	128-bit
Example: 192.168.1.10	Example: 2001:db8::1

4. Why Important?

- Needed for servers, VMs, containers, and cloud networks.
- Helps in routing, security rules, and troubleshooting.
- Common in AWS, Azure, and GCP networking.



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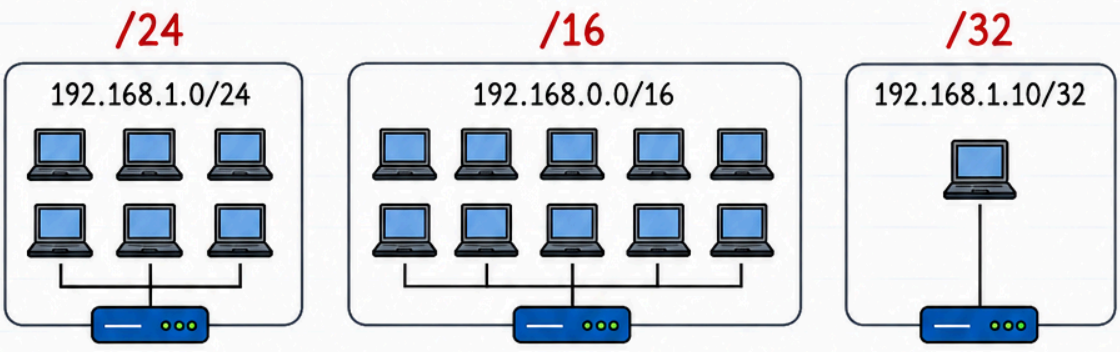
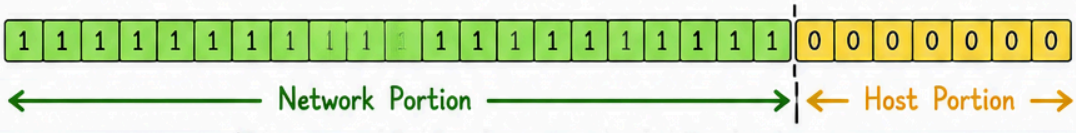
4. CIDR (Classless Inter-Domain Routing)





/24 means 24 bits for network, 8 bits for hosts

CIDR tells how many bits of an IP address belong to the network.
It is used to define network size and is heavily used in cloud platforms such as AWS VPC.

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1. What is CIDR?

- CIDR defines the network portion of an IP range.
- Written using slash notation.
- Example: 192.168.1.0/24

2. Common Examples

CIDR Notation	Number of IP Addresses
/24	= 256 IPs
/16	= 65,536 IPs
/32	= Single IP

3. Why Useful?

Efficient IP allocation.

Better routing and subnet planning.

Common in VPC and cloud design.

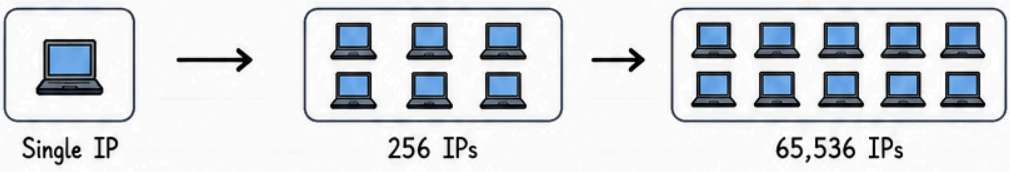
4. Example

192.168.1.0/24

192 . 168 . 1 . 0

24 bits = network

8 bits = hosts



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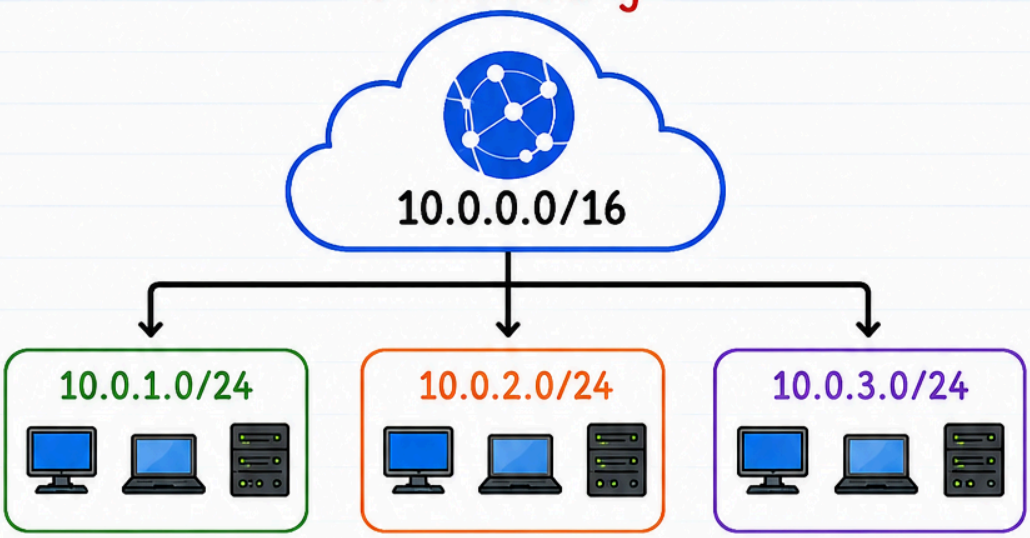
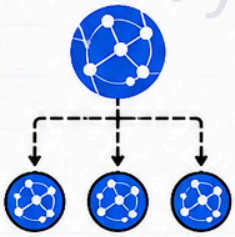
5. Subnetting



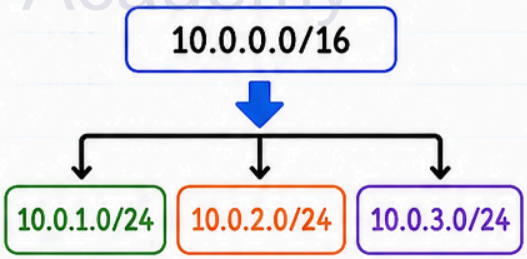
Subnetting means dividing a large network into smaller networks. It improves organization, security, and performance, and is very common in cloud networking.

1. What is Subnetting?



- Dividing a large network into smaller networks.
- Makes networks easier to manage.

2. Example



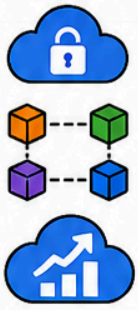
3. Benefits

Better Security

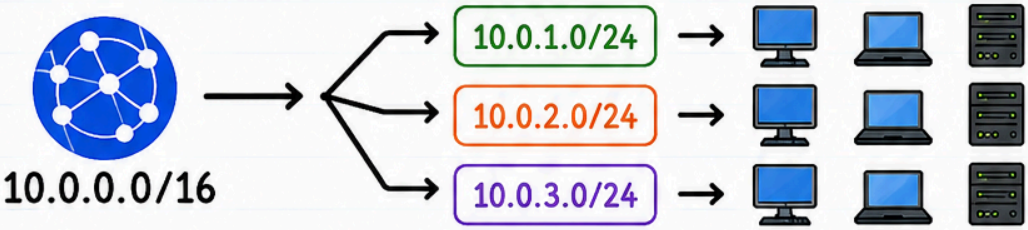
Better Performance

Easy Management

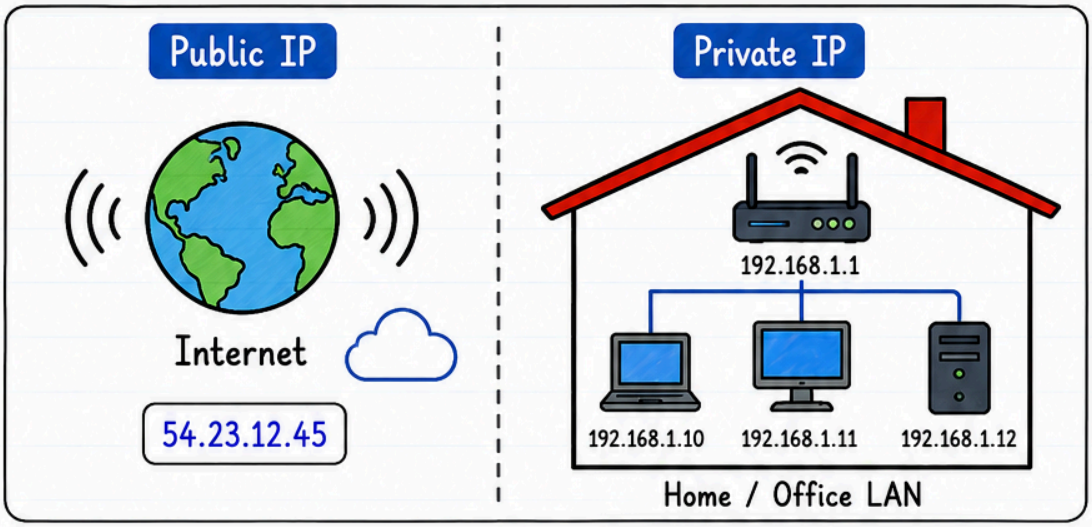
4. DevOps / Cloud Use



- Public and private subnets in cloud.
- Isolation for apps, databases, and services.
- Useful for VPC design and scaling.



6. Public vs Private IP



IP addresses can be public or private. Public IPs are reachable over the Internet, while private IPs are used inside internal networks such as homes, offices, and cloud VPCs.

1. Public IP

- Accessible from the Internet.
- Globally unique.
- Example: 54.23.12.45

2. Private IP

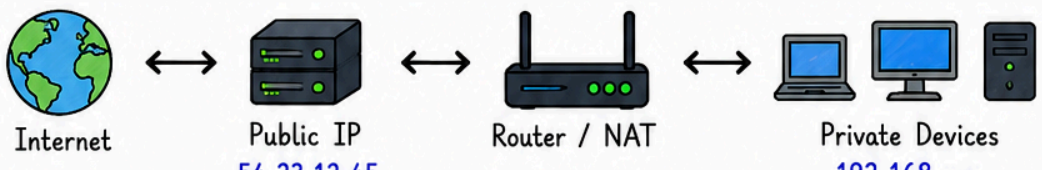
- Used inside private networks.
- Not directly reachable from the Internet.
- Common in LANs and VPCs.

3. Private IP Ranges

- 10.0.0.0/8
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0/16

4. Key Difference

- Public = Internet-facing.
- Private = internal communication.
- NAT is often used to connect private networks to the Internet.



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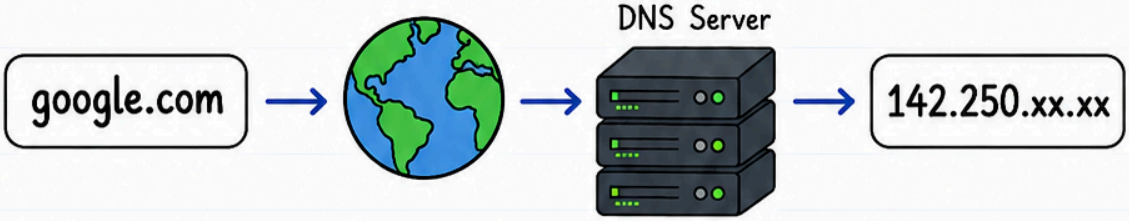
7. DNS (Domain Name System)



DNS converts human-friendly domain names into IP addresses.
It acts like the Internet phonebook so users can access websites using names instead of numeric addresses.

1. What is DNS?

- Converts domain names to IP addresses.
- Helps browsers find servers.

2. Example

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3. Port

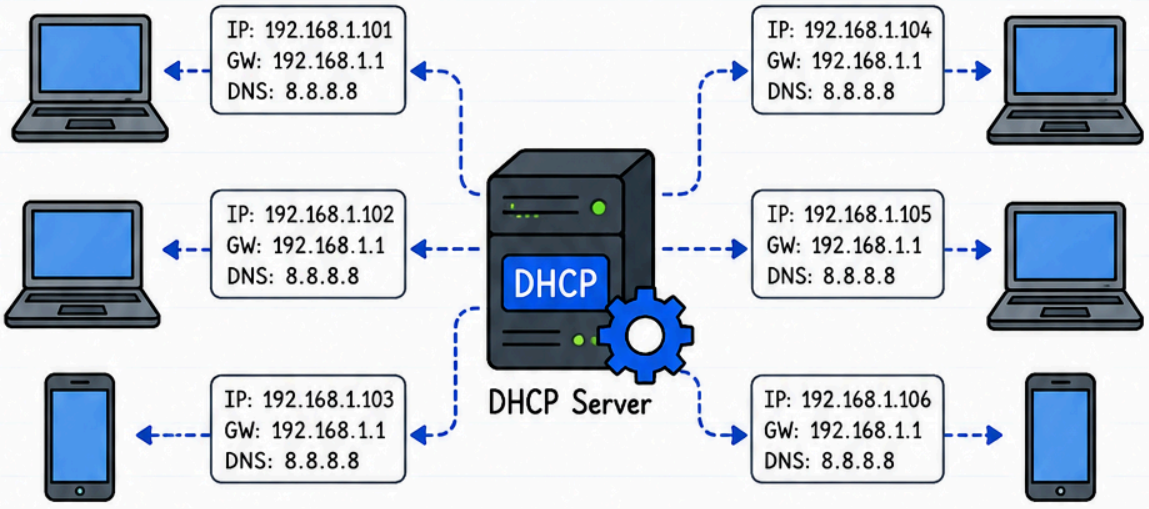
DNS Port

4. Why Important?

- Easy access to websites and services.
- Critical for apps, APIs, and cloud systems.
- Used constantly in troubleshooting.

DNS = Internet Phonebook

8. DHCP



DHCP automatically assigns network settings to devices. It saves time and ensures devices receive an IP address, gateway, and DNS configuration without manual setup.

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1. What DHCP Assigns

- IP Address
- Gateway
- DNS

2. Ports

67 / 68

DHCP Ports

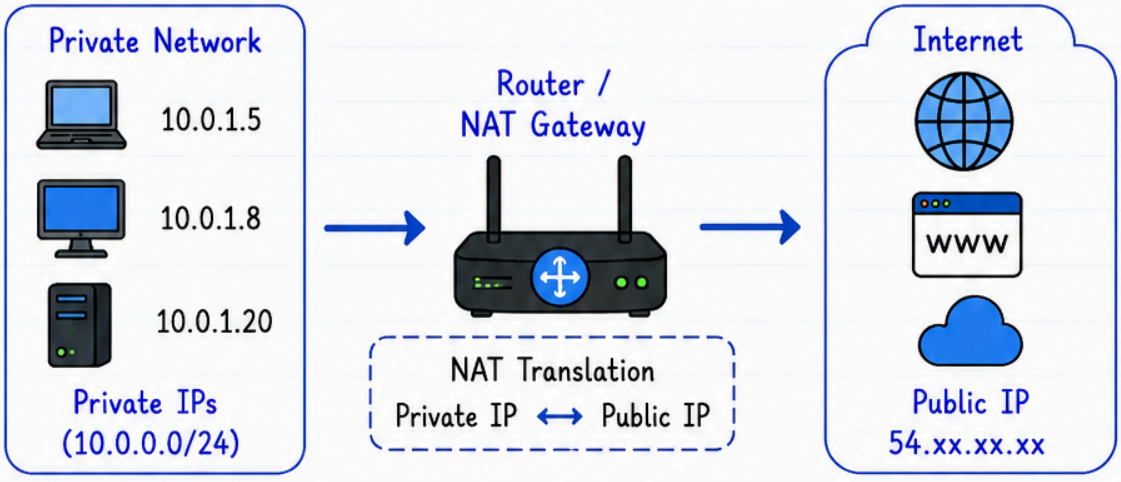
3. How It Helps

- Automatic configuration.
- Reduces manual errors.
- Useful in offices, data centers, and cloud labs.

4. Simple Flow

1 Discover → 2 Offer → 3 Request → 4 Acknowledge

9. NAT (Network Address Translation)



NAT converts private IP addresses to public IP addresses and vice versa. It allows internal systems to access the Internet while conserving public IP addresses.

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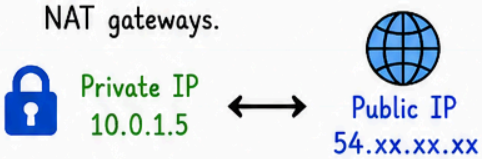
1. What is NAT?

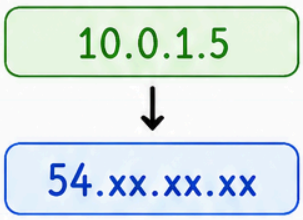
- Converts Private IP ↔ Public IP.
- Common at routers and NAT gateways.



2. Benefits

- Saves Public IPs.
- Gives Internet access to private servers.
- Adds a layer of address abstraction.

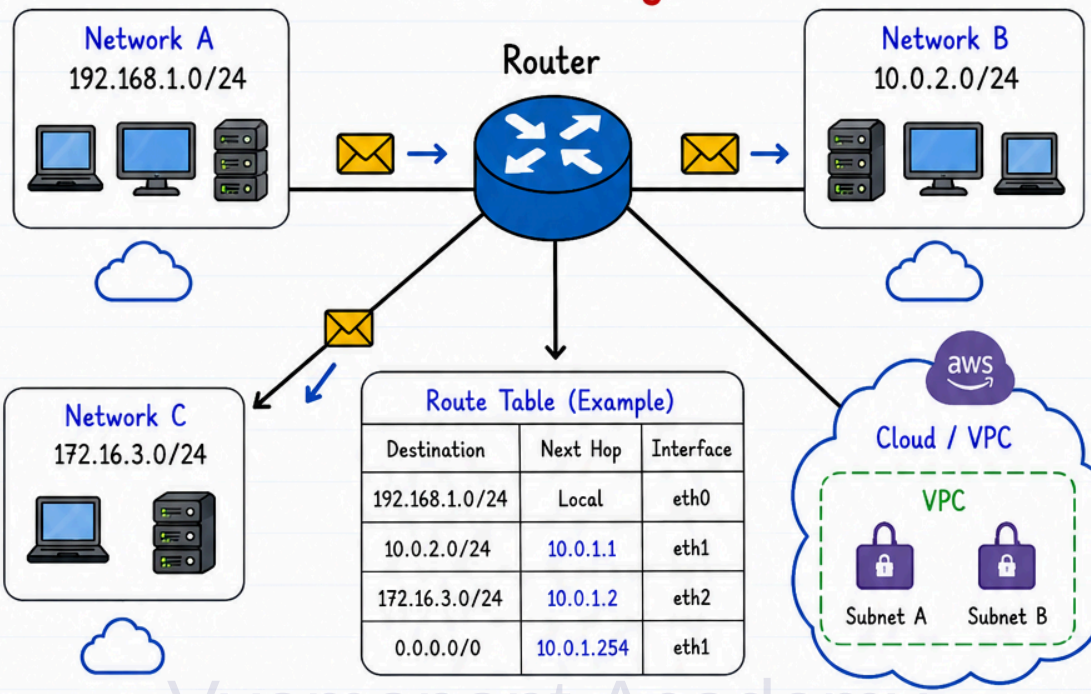
3. Example



4. Where Used?

- Home/office routers.
- Cloud NAT gateways.
- Private subnets in VPCs.

10. Routing



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Routing is the process of moving packets between different networks.
Routers and route tables decide the best path for traffic to reach its destination.

1. What is Routing?

- Moves packets between networks.
- Required when source and destination are on different networks.

2. Done By

Router

Route Tables

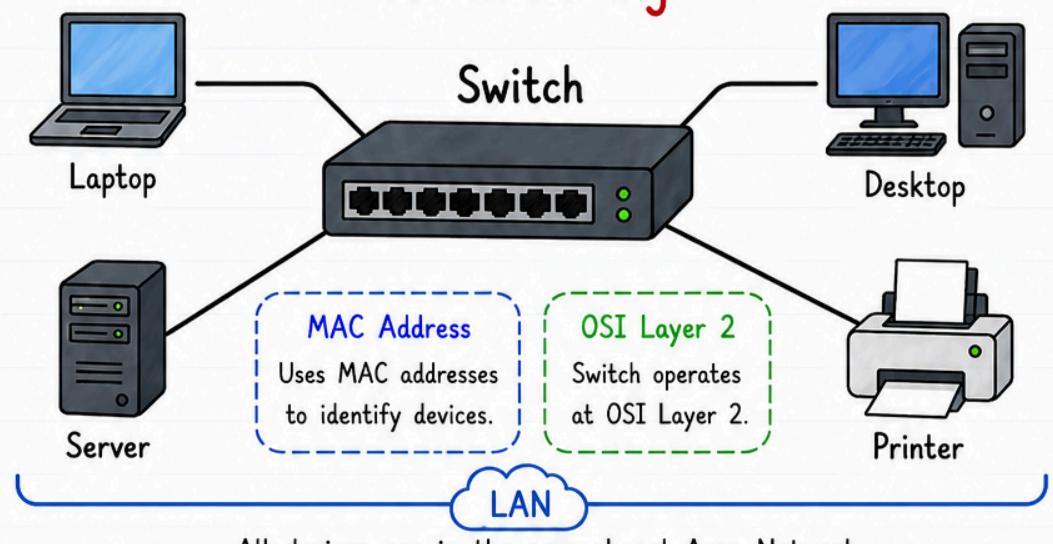
3. How It Works

1. Reads destination IP.
2. Checks route table.
3. Forwards packet to the next hop.

4. Cloud / AWS Use

- AWS uses route tables inside a VPC.
- Directs traffic to Internet Gateway, NAT Gateway, or local targets.
- Very important for subnet communication.

11. Switching



All devices are in the same Local Area Network

Switching connects devices inside the same local area network (LAN).
A switch forwards data using MAC addresses and operates at OSI Layer 2.

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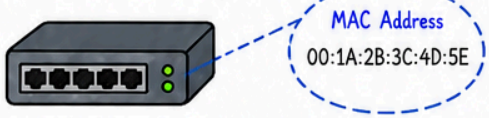
1. What is Switching?

- Connects devices inside the same LAN.
- Uses MAC addresses to identify devices.
- Works at OSI Layer 2.



2. How It Works

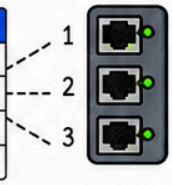
1. Receives a frame
2. Checks MAC table
3. Forwards frame to correct port

3. Key Points

- Fast local communication.
- Common in offices and data centers.
- Unlike routing, it stays inside one network.

MAC Address Table (Example)

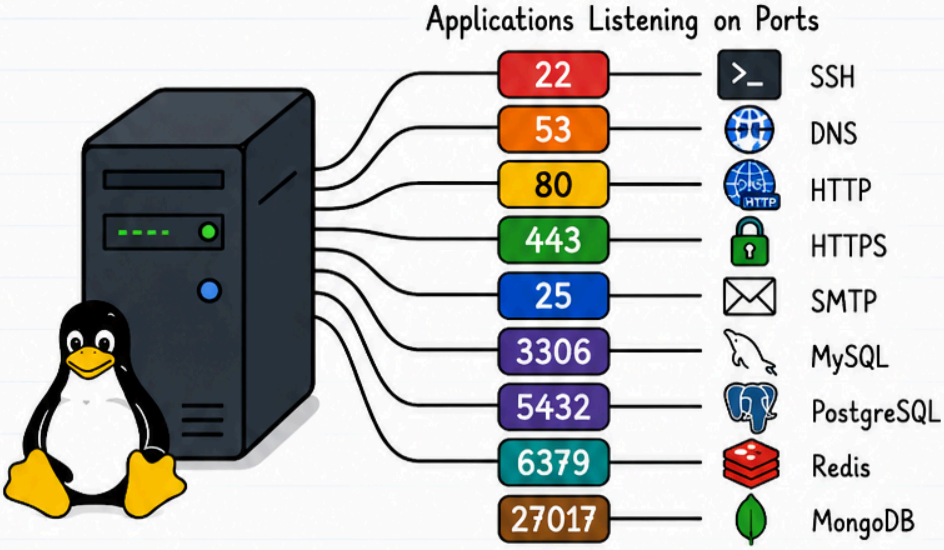
Port	MAC Address
1	00:1A:2B:3C:4D:5E
2	00:1B:2C:3D:4E:5F
3	00:1C:2D:3E:4F:60
...	...



4. Why Important in DevOps?

- Connects servers in racks and data centers.
- Important for local networking and troubleshooting.
- Useful for Kubernetes nodes, VMs, and application servers in a LAN.

12. Ports



Ports are logical numbers used to identify applications and services on a system. They help network traffic reach the correct service.

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1. What are Ports?

- Identify specific applications.
- Used with TCP and UDP.
- Help clients reach the correct service.

2. Common Ports

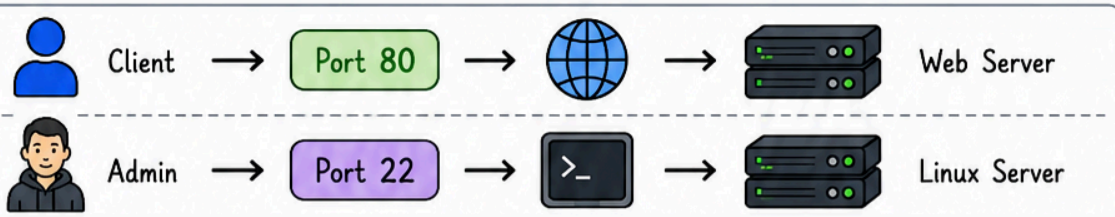
22	= SSH
53	= DNS
80	= HTTP
443	= HTTPS
25	= SMTP

3. Database / DevOps Ports

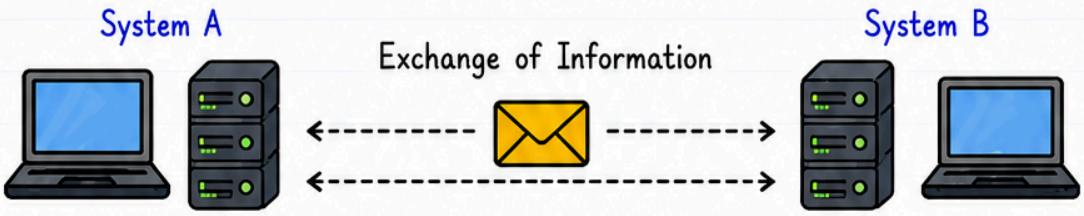
3306	= MySQL
5432	= PostgreSQL
6379	= Redis
27017	= MongoDB

4. Why Important?

- Needed for firewall rules.
- Useful in application deployment and troubleshooting.
- Helps DevOps engineers expose the right services.



13. Protocols



HTTP	HTTPS	TCP	UDP	FTP	SSH	SMTP	DNS
Web	Secure Web	Reliable	Fast	File Transfer	Secure Login	Email Send	Name Lookup

Protocols are rules that define how devices and applications communicate over a network. Different protocols are used for web access, file transfer, remote login, email, and name resolution.

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1. What are Protocols?

- Rules for communication.
- Define data format and behavior.
- Allow devices and apps to understand each other.

2. Common Examples

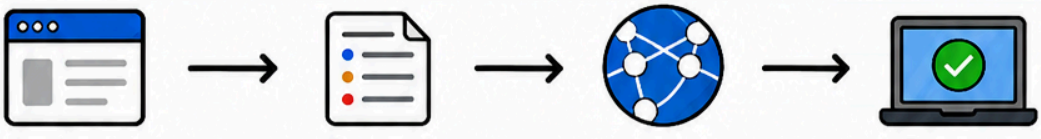
HTTP	= web browsing
HTTPS	= secure web browsing
TCP	= reliable transport
UDP	= fast connectionless transport

3. More Protocols

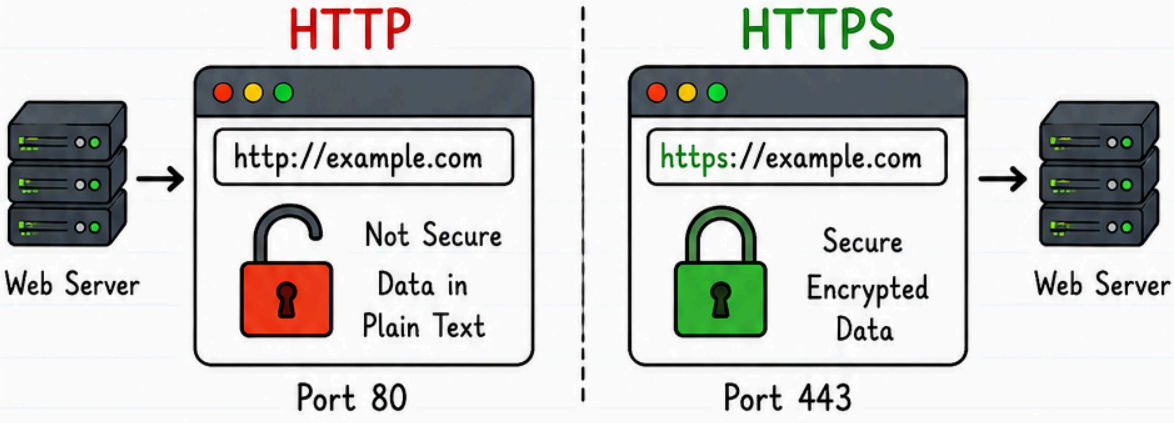
FTP	= file transfer
SSH	= secure remote login
SMTP	= sending email
DNS	= name resolution

4. Why Important?

- Essential for cloud, APIs, and DevOps tools.
- Helps in troubleshooting network issues.
- Each protocol has a specific purpose.



14. HTTP vs HTTPS



HTTP and HTTPS are web protocols used by browsers and servers. HTTPS is the secure version of HTTP because it uses SSL/TLS to protect data in transit.

1. HTTP

- Not secure
- Plain text
- Port 80
- No SSL/TLS

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2. HTTPS

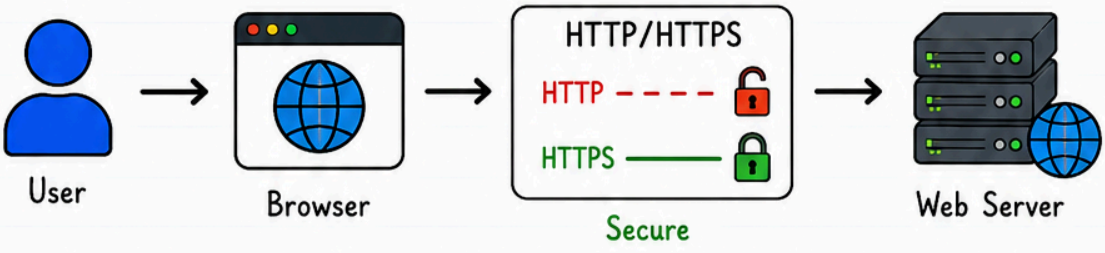
- Secure
- Encrypted
- Port 443
- Uses SSL/TLS

3. Comparison

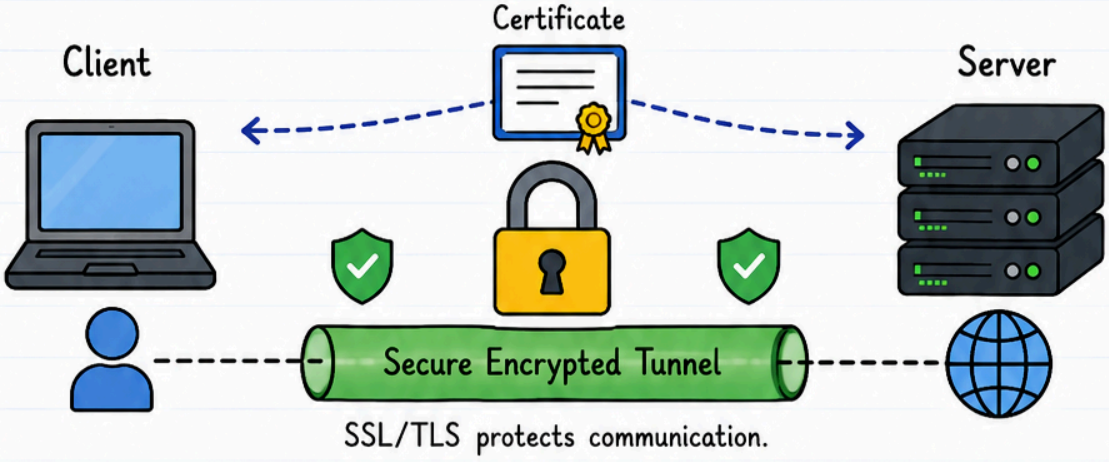
	HTTP	HTTPS
Security	Not Secure	Secure
Data Format	Plain Text	Encrypted
Port	80	443
SSL/TLS	No	Yes

4. Why HTTPS Matters

- Protects passwords and sensitive data.
- Used for websites, APIs, and payments.
- Important for security and trust.



15. SSL/TLS



SSL/TLS provides encryption, authentication, and data integrity for secure communication. It is used by HTTPS and many modern secure applications.

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1. What is SSL/TLS?

- Security protocol for secure communication.
- TLS is the modern replacement for SSL.
- Used widely on the Internet.

2. What It Provides

Encryption
Encrypts data so it cannot be read by others.

Authentication
Verifies the identity of the server.

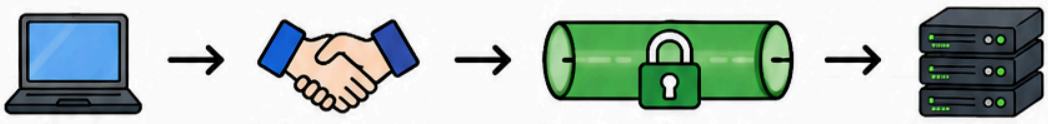
Data Integrity
Ensures data is not changed during transmission.

3. Where Used?

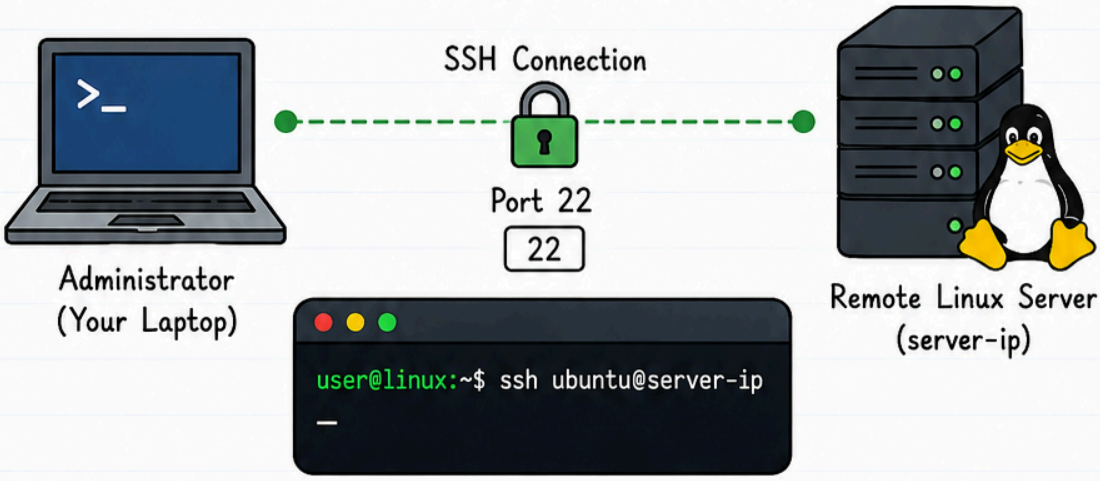
- HTTPS websites
- APIs
- Secure applications and services

4. Key Takeaway

- Keeps data safe in transit.
- Helps verify server identity.
- Important for web and cloud security.



16. SSH (Secure Shell)



SSH is used for secure remote login and command execution on Linux and Unix systems. It is a standard tool for server administration in DevOps.

1. What is SSH?

- Secure remote login protocol.
- Encrypted communication.
- Common for Linux server administration.

2. Port & Example

Port 22

Example Command:

```
ssh ubuntu@server-ip
```

3. Common Uses

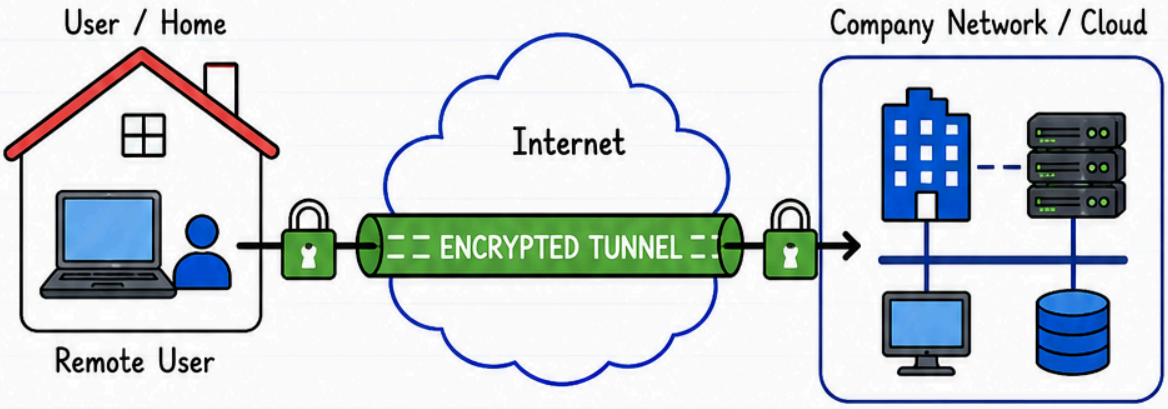
- Remote login
- Running commands
- Managing cloud servers
- Secure file transfer

4. Why Important in DevOps?

- Used daily to manage servers.
- Important for troubleshooting and automation.

Widely used in AWS, Azure, and GCP environments.

17. VPN (Virtual Private Network)



A VPN creates an encrypted tunnel over the Internet. It is used for secure remote access, site-to-site connectivity, and hybrid cloud networking.

1. What is VPN?

- Encrypted tunnel over the Internet.
- Protects traffic between endpoints.
- Used for secure access.

2. Common Uses

- Secure remote access
- Site-to-site connectivity
- Hybrid cloud networking

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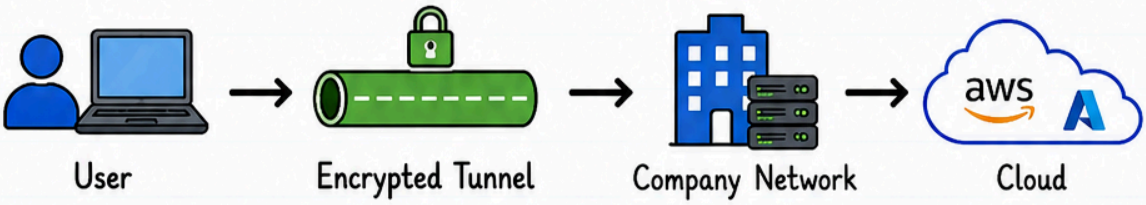
3. Benefits

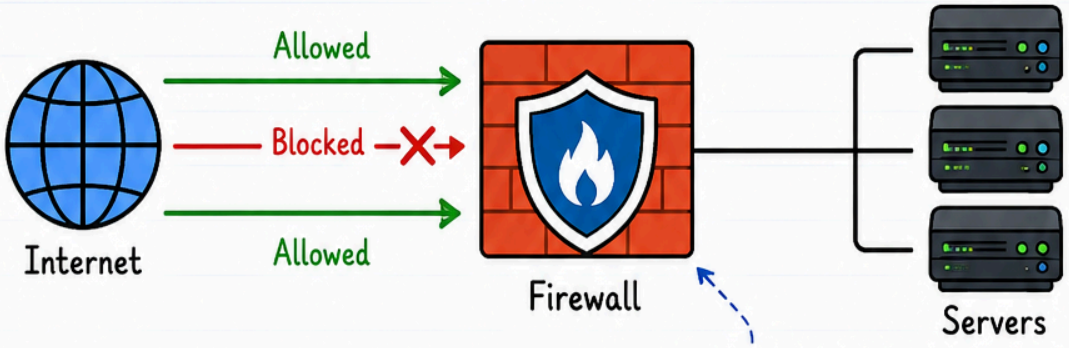
- Privacy
- Encryption
- Secure access from anywhere

4. Why Important?

- Used by remote teams.
- Helps connect offices and cloud networks.
- Important for secure administration and access.



18. Firewalls



Firewalls control incoming and outgoing traffic. They allow or block traffic based on rules such as port, IP address, and protocol.

- Port (e.g., 22, 80, 443)
- IP Address (e.g., 192.168.1.10)
- Protocol (e.g., TCP, UDP, ICMP)

1. What is a Firewall?

- Security control for network traffic.
- Filters incoming and outgoing connections.
- Protects systems and services.

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2. Rule Basics

- Port
- IP Address
- Protocol

Action	Source	Port	Protocal
ALLOW	Any	22	TCP
ALLOW	Any	443	TCP
BLOCK	Any	Any	Any
Default: Block unknown traffic			

3. AWS Examples

- Security Group = stateful
- Network ACL = stateless

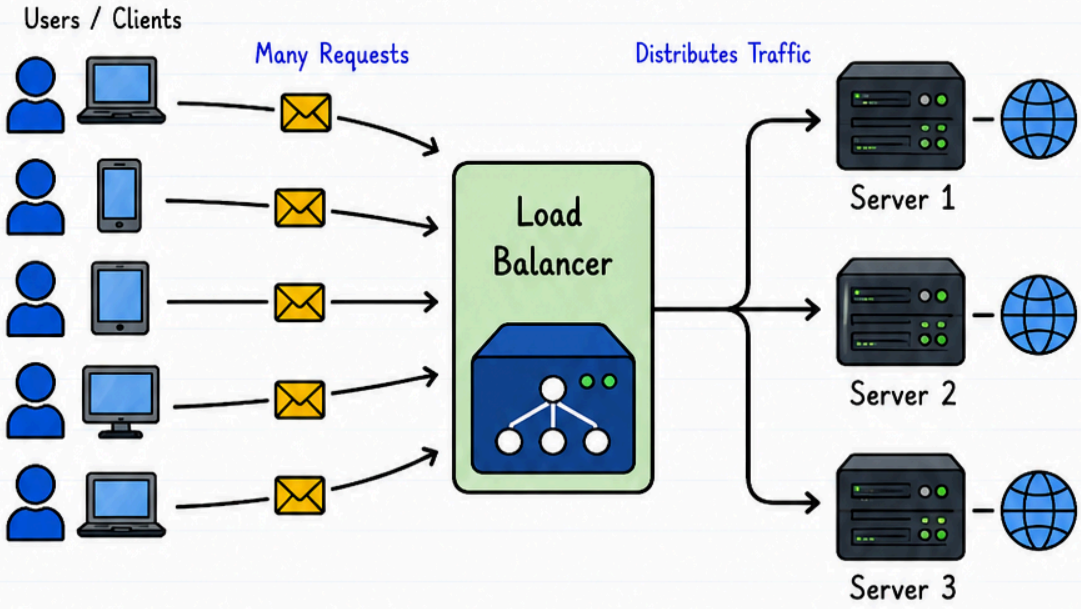
Security Group (Stateful)

Network ACL (Stateless)

4. Why Important?

- Protects servers and applications.
- Helps enforce least privilege.
- Critical for cloud and DevOps security.

19. Load Balancer



A load balancer distributes incoming traffic across multiple servers.
It improves availability, fault tolerance, and scalability.

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1. What is a Load Balancer?

- Distributes traffic across servers.
- Acts between clients and backend systems.
- Helps balance workload.

2. Benefits

- High Availability
- Fault Tolerance
- Scalability

3. Types

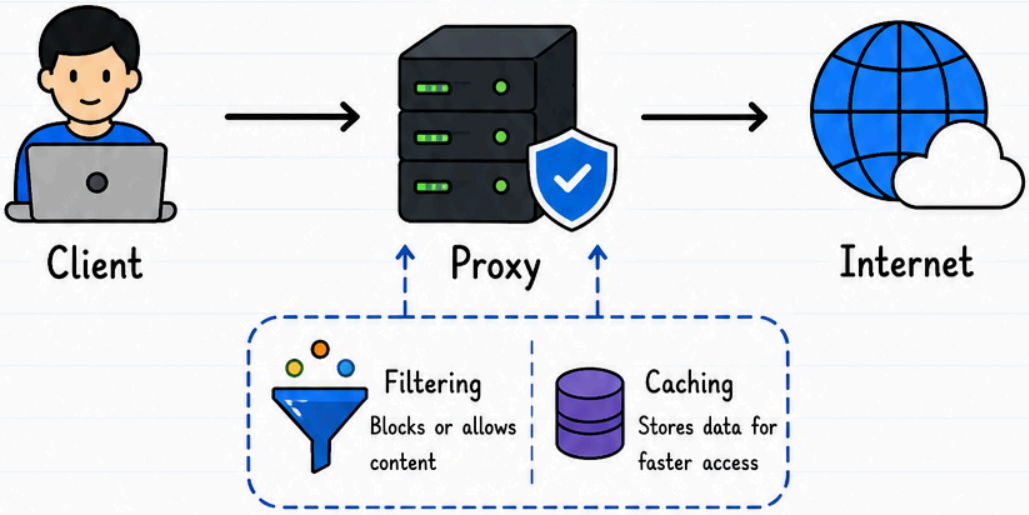
L4 Layer 4 = TCP/UDP

L7 Layer 7 = HTTP/HTTPS

4. Why Important in Cloud?

- Used for web apps and APIs.
- Supports growth and reliability.
- Common in cloud platforms and Kubernetes environments.

20. Proxy



A proxy sits between a client and the Internet. It can hide the client IP, filter content, and cache data for improved control and performance.

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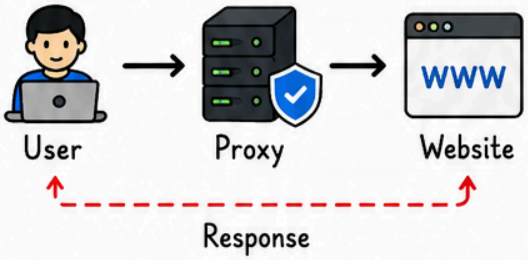
1. What is a Proxy?

- Middle layer between client and Internet.
- Represents the client.
- Used to control outbound access.

2. Purpose

- Hide client IP
- Content filtering
- Caching

3. Simple Example



4. Key Takeaway

- Proxy manages client-side traffic.
- Common in enterprises and secure networks.
- Useful for access control and performance.